

AIML Professional Portfolio Artifact

Machine Learning Algorithm Visual Framework

From Tabular Data to Generative AI

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1. Title

Machine Learning Algorithm Framework: From Tabular Data to Generative AI

2. Introduction

Machine learning algorithms are the foundation of modern artificial intelligence systems. They allow computers to identify patterns, make predictions, classify information, group similar data, and generate new content. However, not every algorithm is useful for every problem. The best algorithm depends on the type of data, the task, and whether the available data includes labels.

This artifact presents a clear visual framework for understanding major machine learning algorithms. It organizes algorithms by learning type, application domain, example use case, and brief explanation. The goal is to make algorithm selection easier to understand and to show how different models apply across tabular data, computer vision, natural language processing, and generative AI.

3. Description

This portfolio artifact contains a visual infographic and supporting explanation. The infographic groups machine learning algorithms into major categories and shows how they relate to different AI/ML application domains.

The framework includes the following algorithms:

- Linear Regression
- Logistic Regression
- Decision Tree
- Random Forest
- Gradient Boosting
- K-means Clustering
- Principal Component Analysis
- K-Nearest Neighbors
- Convolutional Neural Network
- Vision Transformer
- BERT
- GPT
- Generative Adversarial Network
- Diffusion Model
- Large Language Model

Each algorithm is connected to its learning type, application domain, example use case, and simple explanation.

4. Objective

The objective of this artifact is to:

- Clearly identify and categorize major machine learning algorithms.
- Explain whether each algorithm is supervised, unsupervised, deep learning-based, or generative.
- Connect each algorithm to relevant domains such as tabular data, computer vision, NLP, and generative AI.
- Provide practical real-world examples.
- Communicate AI/ML concepts in a clear and professional way.
- Demonstrate understanding through a portfolio-ready artifact.

5. Process

The artifact was created using the following process:

1. **Reviewed the assignment instructions:** The assignment required a visual framework containing at least 8–10 machine learning algorithms.
2. **Identified the main categories:** Algorithms were grouped by learning type and application domain.
3. **Selected representative algorithms:** Common algorithms were chosen from supervised learning, unsupervised learning, computer vision, NLP, and generative AI.
4. **Matched algorithms to use cases:** Each algorithm was connected to a real-world example.
5. **Created brief explanations:** Each algorithm was explained in beginner-friendly language.
6. **Designed the visual framework:** The infographic was created in LaTeX using professional formatting and visual grouping.
7. **Prepared the portfolio page:** The artifact was organized with the required portfolio sections, including objective, process, tools, value proposition, unique value, relevance, and references.

6. Algorithm Classification Table

Algorithm		Type	Domain		Use Case	Brief Explanation
Linear Regression		Supervised	Tabular Data		Predicting house prices or sales revenue	Learns a straight-line relationship between input features and a numerical output.
Logistic Regression		Supervised	Tabular	Data, NLP	Spam detection, risk classification	Predicts categories such as yes/no, true/false, or high/low risk.
Decision Tree		Supervised	Tabular Data		Loan approval, fruit classification, medical triage	Uses if/then questions to classify or predict outcomes.
Random Forest		Supervised	Tabular Data		Fraud detection, credit scoring, customer churn prediction	Combines many decision trees to improve accuracy and reduce overfitting.
Gradient Boosting		Supervised	Tabular Data		Customer churn prediction, ranking, risk modeling	Builds models step by step so each new model corrects earlier mistakes.
K-means Clustering		Unsupervised	Tabular Data		Customer segmentation, product grouping	Groups similar data points into clusters without using known labels.
Principal Component Analysis		Unsupervised	Tabular	Data, Preprocessing	Data reduction, visualization	Reduces many features into fewer important components while keeping useful patterns.
K-Nearest Neighbors		Supervised / Similarity-Based	Tabular	Data, Recommendation	Product recommendation, simple classification	Classifies a new example by comparing it to nearby similar examples.
Convolutional Neural Network		Deep Learning / Supervised	Computer Vision		Image classification, object detection, medical imaging	Learns visual patterns such as edges, shapes, textures, and objects.

Algorithm		Type	Domain		Use Case	Brief Explanation
Vision Transformer	Trans-	Deep Learning / Supervised	Computer Vision	Vi-	Image classification, image understanding	Uses attention to understand relationships between parts of an image.
BERT		Self-Supervised / NLP	Natural Language Processing	Lan- Process-	Sentiment analysis, text classification, question answering	Understands text by learning context from both directions in a sentence.
GPT		Self-Supervised / Generative AI	NLP, Generative AI		Chatbots, summarization, text generation	Uses transformer architecture to predict and generate language.
GAN		Generative AI	Computer Vision, Synthetic Data	Vi-	Image generation, synthetic training data	Uses two neural networks: one generates content and the other evaluates whether it looks real.
Diffusion Model		Generative AI	Image Generation, AI	Genera- Creative	Text-to-image generation, image editing	Creates images by starting with noise and gradually denoising it into a realistic result.
Large Language Model	Language	Generative AI	NLP, Generative AI		Chatbots, writing support, coding assistance	Generates text by learning language patterns from large-scale text data.

Table 1: Classification of machine learning algorithms by type, domain, use case, and explanation.

7. Explanation of the Classification Process

The algorithms were classified using three main criteria: learning type, application domain, and practical use case.

7.1. Learning Type

The first classification criterion was the learning type.

- **Supervised learning** algorithms use labeled data. This means the correct answer is known during training.
- **Unsupervised learning** algorithms do not use labels. They discover patterns, groups, or simplified representations.
- **Deep learning** algorithms use neural networks with many layers to learn complex patterns from data.
- **Generative AI** models create new content such as text, images, or synthetic data.

7.2. Application Domain

The second classification criterion was the application domain.

- **Tabular Data:** Data arranged in rows and columns, such as spreadsheets and databases.
- **Computer Vision:** Image and video data.
- **Natural Language Processing:** Text and language data.
- **Generative AI:** Systems that create new content.

7.3. Use Case

The third criterion was the practical task each algorithm supports. For example, Linear Regression can predict numerical values, K-means can group customers, CNNs can classify images, and GPT models can generate text.

This classification process shows that algorithm selection depends on the type of data, the availability of labels, and the final goal of the task.